

Final Project Proposal

1. Certificate Name:

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2: Project title and description

Title: Online shoppers purchasing intention and behavior profiling Api

This dataset is used to build a classifier that predicts whether a user will complete a purchase transaction is highly for ecommerce conversation rate optimization . This project implements a comprehensive machine learning classification pipeline to predict a visitor's purchasing intent(Revenue: TRUE OR FALSE) using their behavioral session data. To optimize predictions, an unsupervised clustering .

3: Problem type

classification (supervised) coupled with clustering(unsupervised feature engineering.

the ultimate target is a binary classification output Revenue true or false .

4: Dataset

source : online shoppers purchasing intention dataset hosted on Kaggle

size: 12,330 rows and 18 attribute .

target: Revenue (binary true if a purchase was made , otherwise false) .

Key Columns:

- Administrative, Informational, Product Related: Number of pages visited in each category.
- Duration: Time spent in each category.
- Bounce Rates, Exit Rates: Metrics measuring user engagement and site departure.
- Page Values: The average value of the page viewed before the user completed a transaction.
- Month, Region, Browser: Categorical data about the user.
- Revenue: The target label (TRUE = Purchase, FALSE = No Purchase).

Preprocessing plan: apply one-hot encoding to categorical session attributes normalize ranges with standardscaler and minmaxscaler, and execute an 80/20 train/test split.

5: Algorithms i plan to train

the architecture utilizes a combination of advanced clustering and multiple distinct classification algorithms.

- K-means used unsupervised feature engineering step to label users into browsing behavior profiles before classification.
- Gaussian Mixture Model(GMM) Evaluated against k-means to identify if soft-clustering probability features enhance downstream classifier score.
- Logistic Regression the baseline classification model used for benchmark metrics.

- Random Forest Classifier a robust ensemble method to interpret complex non linear combination of site interactions.
- Gradient Boosting classifier a sequential boosting technique used to minimize bias and boost accuracy on the tabular data.
- XGboosting classifier: an independent research algorithm built to extract maximum predictive precision from multi-feature retail environments.

6: Evaluation plan

Metrics for all models: Accuracy, Precision, Recall, and F1-Score

Best-model rule: Models will be ranked and selected based on the highest F1-Score on the validation split

The F1-score balances True Approvals vs False alarms, which is crucial since buying customers are typically a minority class in e-commerce traffic datasets

7. Deployment Sketch

Framework: FastAPI

Endpoint: `POST /predict`

input json example:

```
json
{
  "Administrative": 54.2,
  "Duration": 0.0,
  "Product Related": 650.5,
  "Bounce Rates": 0.0,
  "Exit Rates": 0.02,
  "Page Values": 34.8,
  "Month": "May",
  "Revenue": "True or False"
}
```

output json example:

```
json
{
```

```
"purchase_intention": "True",  
"probability": 0.89,  
"user_behavior_cluster": 2  
}
```

6: Repository plan

online-shopper-intent-api/

├── dataset/

| ├── **shopping.csv**

├── src/

| ├── preprocess.py # Session clustering and feature engineering

| ├── train.py # Training pipeline for Logistic, RF, Gradient Boosting, XGBoost

├── api/

| ├── app.py # FastAPI framework serving predictions

├── models/

| ├── best_classifier.pkl

| ├── behavior_clusterer.pkl

| └── scaler.pkl

├── README.md # Installation and local deployment guide

├── requirements.txt # Project constraints

└── project_paper.md